

DEPARTMENT OF INDUSTRIAL & MANAGEMENT ENGINEERING

DMS201: INTRODUCTION TO OPERATIONS MANAGEMENT

QUIZ # 01 |2025-2026, SEMESTER III|

NOTE THE FOLLOWING

- 01) Total number of questions is **THIRTY (30)**.
- 02) Questions are: True/False; Range Based (Qualitative & Quantitative), MCQ (FIVE (05) options with ONE (01) or MORE CORRECT answer(s)).
- 03) Marks: CORRECT is +2/Plus TWO; WRONG is -1/Minus ONE; No ANSWER is 0
- 04) Time duration is FORTY FIVE (45) / SIXTY (60) minutes for non-DAP / DAP students.
- 05) Closed Notes/Book examination.

Assume Demand is 20 units per 20 minutes while Capacity (ability to process) per machine is 40 units per 30 minutes. With this information we can conclude that flow rate is 80 units per hours?rrr

False

Given the data that demand is 20 units per 20 minutes while Capacity (ability to process) per machine is 40 units per 30 minutes, we have

Demand = $20 \times 3 = 60$ units per hour

Capacity = $40 \times 2 = 80$ units per hour

Flow rate = $\text{Min}(\text{Demand}, \text{Capacity}) = \text{Min}(60, 80) = 60$ units per hour

Processing Time = $\{m/\text{Capacity}\}$

True

Fifteen equally skilled electricians fit compressor and rotating parts in air conditioners (A.C.) in the shop-floor of VOLTAS. All the fifteen workers complete the fitting of 240 number of A.Cs in an 8 hour shift. It is also given that the value of output is Rs. 12,00,000/- while labour and electricity costs are Rs. 3,25,000/- and Rs. 1,78,125/- respectively. Then find the ratio of multi-factor productivity wrt labour productivity is?

A 1.20 hour/A.C.

B 1.30 hour/A.C.

C 1.40 hour/A.C.

D 1.50 hour/A.C.

E 1.70 hour/A.C.

Given the information we have

- Labour productivity = $240 \text{ A.C.} / (15 \text{ workers} \times 8 \text{ hours}) = 2.00 \text{ A.C./hour}$.

- Multi-factor productivity = $1200000 / (325000 + 178125) = 2.39$

Hence: Multi-factor productivity/Labour productivity = $(2.39/2.00) = 1.20$ hour/A.C.

Consider a three-step airport security at Frankfurt Flughafen/Airport (FRA), where

- First step ($i=1$): Verifying ID and boarding pass and it takes THIRTY (30) seconds per passenger.
- Second step ($i=2$): Security check and use of metal detector and it takes TWENTY (20) seconds per passenger.
- Third step ($i=3$): Security check for hand luggage through an X-ray machine and it takes FIFTY FIVE (55) seconds per passenger bag.

If there are THIRTY (30) passengers standing in the queue and with no other passengers in front of them, then the total time taken by the security system is?

A	25 minutes
B	28 minutes
C	30 minutes
D	32 minutes
E	45 minutes

Given the information, it takes $(30 + 55)$ seconds = 85 seconds until the FIRST customer is served.

From then onward, we are serving a customer every 55 seconds, and as we have 30 customers to serve, this will take $(30 - 1) \times 55$ seconds = $29 \times 55 = 1595$ seconds

Total time taken $(85 + 1595)$ seconds = $1680/60 = 28$ minutes

80% of products in the production line are good and a random sample of 4 products is chosen. With this information find the approximate probability that at most 1 of them will be found to be a bad item?

A	0.72
B	0.80
C	0.82
D	0.89
E	0.92

$$P(X \leq 1) = P(X=0) + P(X=1) = {}^4C_0 \times 0.8^4 \times 0.2^0 + {}^4C_1 \times 0.8^3 \times 0.2^1 = 0.82$$

Mr. Basavakiran Shivanand Deshmukh is the GM of Bad Quality Products which manufactures laptop bags. With the current machinery set up, 240 lap top bags are made from 1,000 meters of polyvinyl. Mr. Deshmukh purchases 1,000 meters of polyvinyl and each 5 meters requires 2 man hour to process. In case Mr. Deshmukh decides to increase his production to 260 lap top bags from 1,000 meters of polyvinyl using better trained workman-ship then man hours will increase by 5 hours. What will be the impact on productivity which Mr. Deshmukh will face?

A	Increase of 7.76%
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B	Decrease of 7.76%
C	Increase of 9.26%
D	Increase of 6.67%
E	Decrease of 6.67%
Current man hour productivity = $240/((1000/5) \times 2) = 240/400 = 0.60$ lap top bag per man hour. Furthermore, new man hour productivity = $260/(((1000/5) \times 2) + 5) = 260/405 = 0.64$ lap top bag per man hour. Hence increase in productivity is $((0.64 - 0.60)/0.60) \times 100 = 6.67\%$	

Given $X \sim N(\mu, \sigma^2)$, where N denotes Normal Distribution, the total area covered between $\mu - 2\sigma$ to $\mu + 2\sigma$ is?	
A	90.45%
B	93.45%
C	94.45%
D	94.54%
E	95.45%

For a road side (connecting Erode to Coimbatore) eatery, the average preparation times to make 4 Idlis and 1 plate of Sambar by a two different cooks are 1 minutes and 4 minutes respectively. Remember both are working on the specific preparations only. At 8:00 AM in the morning a bus load of 36 passengers (including driver and helper) come to the eatery. Each of the customer orders a plate (4 Idlis & 1 plate of Sambar). How long (in hours) would it take to complete the demand if there are 2 different persons making Idlis and 2 different persons preparing Sambar (answer to 2 places of decimal).	
Range	1.15 to 1.25
- Capacity_1 for Idlis is 1 plate (or 1 customer) per 1 minute. Considering 2 persons prepare Idlis which means 2 plates (or 2 customers) per 1 minute. Hence $60 \times (1/2) = 30$ plates (or 30 customers) per hour - Capacity_2 for Sambar is 1 plate (or 1 customer) per 4 minutes. Considering 2 persons prepare Sambar which means 2 plates (or 2 customers) per 4 minutes. Hence $60 \times (2/4) = 30$ plates (or 30 customers) per hour - Flow Rate = $\text{Min}(\text{Demand}, \text{Capacity}_1, \text{Capacity}_2) = \text{Min}(\infty, 30, 30) = 30$ plates (or 30 customers) per hour. - Hence time take would be $36 \times (1/30) = 1.2$ hours	

There are 3 machines ($i=1,2,3$) in parallel in the shop floor and all of the 3 machines are doing the SAME operation. The average processing times for the 3 machines ($i=1,2,3$) are 10, 12 and 15 minutes per job respectively. The average throughput per 8 hour shift for the shop floor is (answer to 0 place of decimal)?	
Range	119.999 to 120.001

The calculations are shown below

- Machine ($i=1$) on an average operates 6 jobs her hour
 - Machine ($i=2$) on an average operates 5 jobs her hour
 - Machine ($i=3$) on an average operates 4 jobs her hour
- Hence total $8 \times (6+5+4) = 8 \times 15 = 120$ jobs per 8 hour shift

A primary hospital has the capacity to see 40 patients per 1 hour. The Demand rate is, however, only 335 patients per 6 hours. What is the utilization of the hospital (answer to 2 places of decimal)?

Range	0.99 to 1.01
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The calculations are shown below

- Capacity is $6 \times 40 = 240$ patients per 6 hours
- Demand is 335 patients per 6 hours
- Flow Rate = Minimum {Demand, Process Capacity} = $\text{Min}(335, 240) = 240$ patients per 6 hours
- Utilization = {Flow Rate/Capacity} = $(240/240) = 1$

For a departmental store the Flow Rate is 220 customers per hour while Capacity is 100 customers per 1 hour per 1 billing desk, and there are 2 billing desks. Hence the saving (in hours considering 2 places of decimal) by the department store in an 8 hour shift, is?

Range	0.77 to 0.79
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The calculations are shown below

- Cycle Time = $(60 \times 60) / 220 = 16.4$ seconds per customer per hour.
- Capacity = $(60 \times 60) / (2 \times 100) = 18.0$ seconds per customer per hour.

Departmental store is able to handle the as it saves $(18.0 - 16.4) = 1.6$ seconds per customer per hour, hence $\{(1.6 \times 220) / (60 \times 60)\} \times 8$ hours = 0.78 hours in the eight hour shift

It is given that the distribution of order for slow moving items is $X \sim \text{NB}(p=0.5, r=10)$, while distribution of demand for low requirement items is $X \sim \text{P}(\lambda=15)$. Then the value of bull-whip effect.

Range	1.30 to 1.36
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Bullwhip value = $\text{Variance}(\text{Order}) / \text{Variance}(\text{Demand})$. From data we know that order variance = $r \times (1-p) / p^2 = 10 \times 0.5 / (0.5^2) = 200$ and demand variance = $\lambda = 15$, hence Bullwhip value = $200 / 15 = 1.33$

In a production system we have Stage 01, Stage 02, Stage 03, Stage 04 and Stage 05. It is given that the distributions of order/demand between

- Stage 01 to Stage 02 is $X \sim \text{P}(\lambda=10)$
- Stage 02 to Stage 03 is $X \sim \text{P}(\lambda=15)$

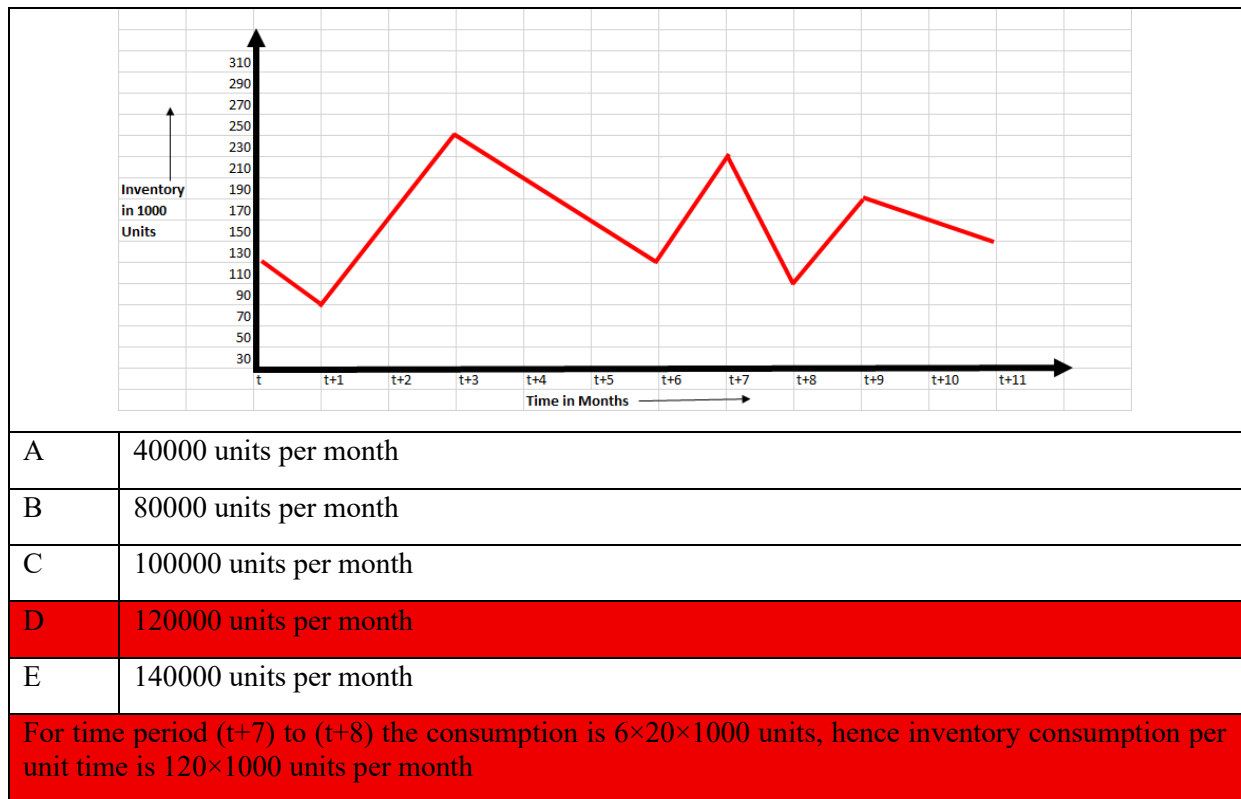
- Stage 03 to Stage 04 is $X \sim P(\lambda=20)$ - Stage 04 to Stage 05 is $X \sim P(\lambda=25)$ With the above information find the ratio of bull-whip effects between Stage 01-Stage 02 & Stage 02-Stage 03 wrt Stage 03-Stage 04 & Stage 04-Stage 05	
Range	0.832 to 0.834
$V(\text{Stage 01-Stage 02})/V(\text{Stage 02-Stage 03}) = 10/15$, while $V(\text{Stage 03-Stage 04})/V(\text{Stage 04-Stage 05}) = 20/25$. Hence ratio is $(10/15)/(20/25) = 0.833$	

Time to finish m units starting with an empty system is given by	
A	Time through an empty process + $[(m - 1) \times \text{Cycle time}]$
B	Time through an empty process + $[\{(m - 1)/m\} \times \text{Cycle time}]$
C	Time through an empty process + $[\{m/(m + 1)\} \times \text{Cycle time}]$
D	Time through a non-empty process + $[(m + 1) \times \text{Cycle time}]$
E	Time through a non-empty process + $[\{(m + 1)/m\} \times \text{Cycle time}]$

The idea where automation for production is implemented using robotics, machines, material handling equipment, storage, etc., is termed as?	
A	Agile Manufacturing (AM)
B	Flexible Manufacturing System (FMS)
C	Computer Numerical Control Manufacturing (CNCM)
D	Computer Integrated Manufacturing (CIM)
E	Computer Aided Design and Manufacturing (CADM)

Important terms/ideas/concepts which is not under the ambit of inventory/stock is?	
A	ABC Analysis, Safety Stock, Back Ordering, Lead time
B	Safety Stock, Reorder point, Inventory Turnover, Just In Time (JIT)
C	Economic Order Quantity (EOQ), Gross Domestic Product, Safety Stock
D	Dead Stock, Lead time, Back Ordering, ABC Analysis
E	Lead time, Inventory Turnover, Economic Order Quantity (EOQ)

From the figure given below, the value for highest inventory consumption per unit time is?
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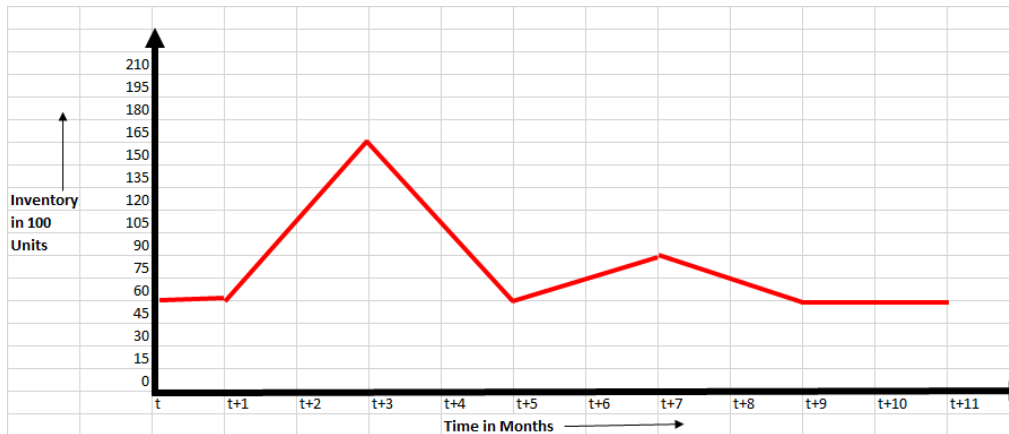
The statistical distribution for Order, when variability of order is high is given by?	
A	Negative Binomial Distribution
B	Poisson Distribution
C	Normal Distribution
D	Geometric Distribution
E	Gamma distribution

Which issue may NOT be a direct factor affecting Supply Chain	
A	Competition
B	E-Commerce
C	Consumer expectations
D	Government Regulation
E	Price of transportation

Given $X \sim NB(0.45, 6)$, where NB denotes Negative Binomial Distribution, the ratio of $E(X)/V(X)$ is?	
A	0.2475
B	0.3209

C	0.4500
D	0.5500
E	0.8181
$E(X) = rq/p$ and $V(X) = rq/p^2$, hence $E(X)/V(X) = [rq/p]/[rq/p^2] = p = 0.45$	

From the figure given below, the average inventory in units for the time period considered is?



A	5448 units per month
B	5484 units per month
C	5844 units per month
D	8454 units per month
E	8544 units per month
Total inventory = $(1/2) \times 4 \times 7 \times 15 \times 100 + (1/2) \times 4 \times 2 \times 15 \times 100 + 4 \times 15 \times 11 = 93000$ Time = 11 months Hence: Average Inventory = $93000/11 = 8454$ units per month	

THIRTY FIVE (35) workers (working with the same efficiency) complete the fittings of washing machines as they are packed for dispatch. The same workers complete the fitting of 225 washing machines in THREE (03) hours. It is given that the value of output is Rs. 20,25,000/- per THREE (03) hour, while labour and electricity costs are Rs. 86,500/- per hour and Rs. 79,725/- per hour respectively. Then find the value of Multi-factor productivity?

A	1.461
B	2.061
C	3.246
D	3.416
E	4.061
Multi-factor productivity = $(2025000/3)/(86500 + 79725) = 4.061$	

A shipping company operates 30 container ships of three (03) capacities (each capacity is 10 in number) values of which are 25,000, 35,000, 55,000 containers between ports. The company wants to load and unload 50,000 containers each week. Assume all the ships of the three varieties always travel fully loaded. The longest average travel time (**Flow Time**) between ports that allows the company to meet their goal of 50,000 containers per week is?

A	2.3
B	2.5
C	2.7
D	3.0
E	3.2

On average there are $10 \times 25000 + 10 \times 35000 + 10 \times 55000 = 1,15,000$ containers transported between ports, and the company wants to load/unload 50,000 containers each time.

Hence Flow Time = $115000/50000 = 2.3$ weeks

During the course of an 3 hour window, there are, a total of 24 students who visit the UG section, DoAA office at IIT Kanpur, and each student spends, on average, 4 minutes with their respective queries. At what rate do students go to the UG section DoAA office at IIT Kanpur (in students per hour)?

Range	119 to 120
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The calculations are shown below

Average inventory, $I = 24/3 = 8$ students per hour

Average flow time 4 minutes, $T = 4/60$ hour

Rate of student flow = $I/T = 8 \times 60/4 = 120$ students per hour

Consider a FIVE (05) stage production system, where

- **First stage ($i=1$):** ONE (01) unit of **item # 01** is processed in THREE (03) minutes.
- **Second stage ($i=2$):** ONE (01) unit of **item # 02** is processed in TWO (02) minutes.
- **Third stage ($i=3$):** ONE (01) unit of **item # 03** is processed in SIX (06) minutes, whereby ONE (01) unit of **item # 03** requires: ONE (01) unit of **item # 01** (obtained from First stage ($i=1$)) and TWO (02) units of **item # 02** (obtained from Second stage ($i=2$)).
- **Fourth stage ($i=4$):** ONE (01) unit of **item # 04** is processed in FOUR (04) minutes.
- **Fifth stage ($i=5$):** ONE (01) unit of **item # 05** is processed in ONE (01) minute, whereby ONE (01) unit of **item # 05** requires: TWO (02) unit of **item # 03** (obtained from Third stage ($i=3$)) and THREE (03) units of **item # 04** (obtained from Fourth stage ($i=4$)).

Find the **total time** (in minutes) to produce a batch of TEN (10) units of **item # 05**?

Range	209 to 211
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The calculations are shown below

To produce ONE (01) unit of Item # 05 we required $1 \times 20 + 01 = 21$ minutes, and the general formulae is $i \times 20 + 01 = 21$ here $i = \text{Number of Item \# 05 produced}$, hence to produce TEN (10) units of Item # 05 we required $10 \times 20 + 01 = 210$ minutes, here $i=10$

We can draw the GANTT chart to visualize the problem and it is as follows

Time	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Item # 01	1	1	1								2	2	2								
Item # 02	1	1	2	2							3	3	4	4							
Item # 03					1	1	1	1	1	1					2	2	2	2	2	2	
Item # 04									1	1	1	1	2	2	2	2	3	3	3	3	
Item # 05																					1

SBI offers customized loans. Customers have specific loan request and when they visit the bank are able to obtain a response on an average within 30 minutes. Queries for SBI loan includes five activities that must be conducted in the sequence described below.

- **Activity 1:** Queries related to why loan is required by a person and is answered in 4 minutes.
- **Activity 2:** The SBI counter person gathers all the related information from the customer on an average in 5 minutes.
- **Activity 3:** The SBI counter person analyzes the information and checks the papers/documents, etc. of the customer to decide the loan and amount and on an average it takes 7 minutes.
- **Activity 4:** The SBI counter person finally checks and verifies all the papers/documents and on an average it takes 2 minutes.
- **Activity 5:** Finally the customer signs the documents and has the loan money transferred to his/her account and on an average it takes 4 minutes.

The whole process is conducted by three (03) SBI persons in a worker-paced line. The assignment of tasks to SBI personnel is as follows: **W1 does activity 1, W2 does activities 2 and 3, and W3 does activities 4 and 5.** You can assume that there exists unlimited demand and that loans are only entering the process at the rate of the bottleneck. Then what is the value of bottleneck (in minutes) in the process ?

Range | 11 to 13

The calculations are shown below

We first compare the capacity levels across the three workers; they are 1/4 loan per minute, 1/12 loan per minute, and 1/6 loan per minute, respectively; so W2, with a processing time of 12 minutes per loan (5 + 7), is the bottleneck.

Assume Demand is 30 units per 20 minutes while Capacity per machine is 40 units per 30 minutes, and there are 5 machines. Find the flow rate/throughput.

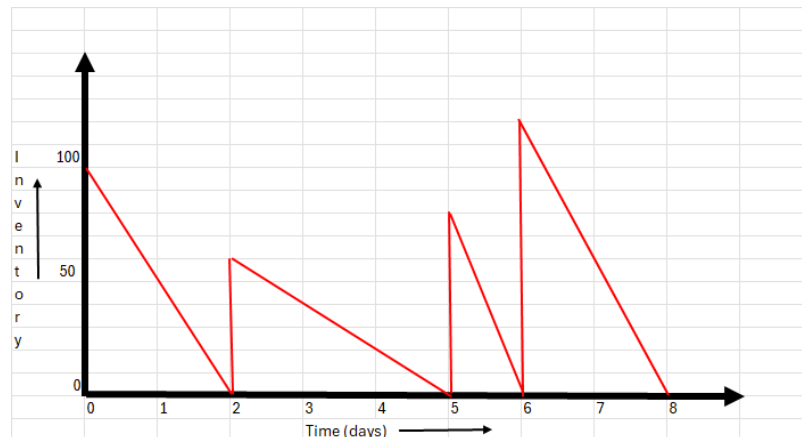
Range | 59 to 61

The calculations are shown below

Thus Demand is $30 \times 3 = 90$ units per $20 \times 3 = 60$ minutes (1 hour) while Capacity (ability to process) per machine is $40 \times 2 = 80$ units per $30 \times 2 = 60$ minutes (1 hour).

Hence Flow Rate or Throughput = $\min(90 \times 3, 80 \times 5) = \min(90, 400) = 90$ units per hour, and it is a Demand Constrained process

From the Inventory versus time graph, one can conclude that the average inventory for the EIGHT (08) days is?



A 73.45

B 54.37

C 57.35

D 43.75

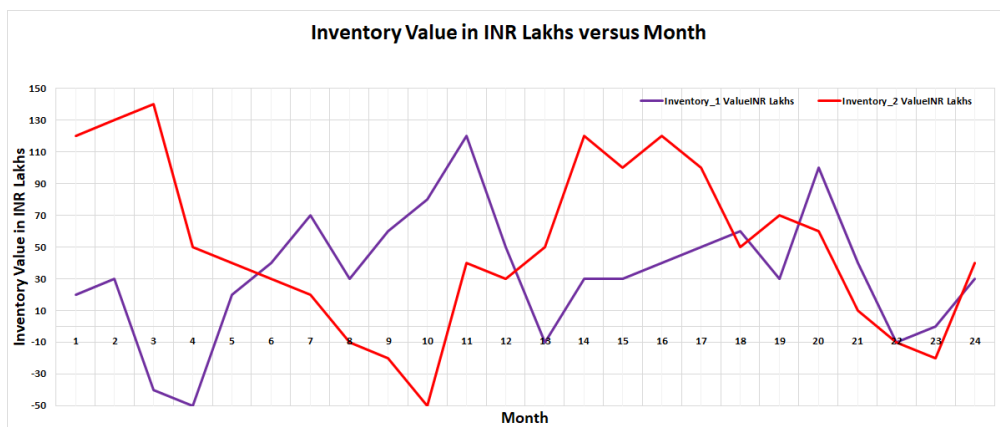
E 34.57

Total inventory = $(1/2) \times 100 \times 2 + (1/2) \times 60 \times 3 + (1/2) \times 80 \times 1 + (1/2) \times 120 \times 2 = 700/2$

Time = 08 days

Hence: Average Inventory = $350/8 = 43.75$

From the graph given we can conclude that the maximum value of Inventory (**combining Inventory 01 and Inventory 02**) and given in INR Lakhs is (answer to 1 place of decimal)?



Range 155 to 165

A rough calculation will give the answer

For a departmental store the **Flow Rate** or **Throughput** is 350 customers per hour while Capacity is 200 customers per 1 hour per billing desk. If departmental store is open from 1000 hours to 1800

hours, then the savings in minutes per day with FIVE (05) billing desks is? Consider all the FIVE (05) billing desks have the same efficiency and all are working continuously without any stoppage.

Range	311 to 313
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For a departmental store the Flow Rate or Throughput is 350 customers per hour while Capacity is 200 customers per 1 hour per billing desk.

The Cycle Time = $1/350 = 0.002857$ hour per customer = $0.002857 \times 60 \times 60 (= 10.2857)$ seconds per customer.

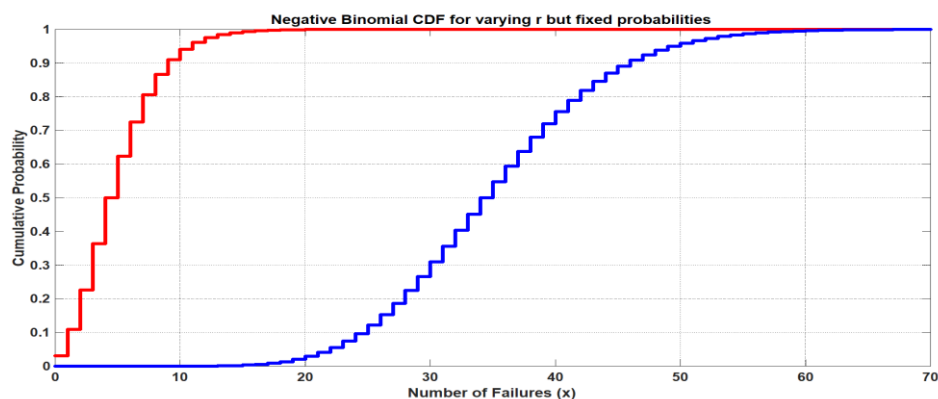
With Capacity we have = $(60 \times 60)/200 (= 18)$ seconds per customer.

With 5 billing desk Capacity = $\{(60 \times 60)/(200 \times 5)\} (= 3.6)$ seconds per customer.

5 Billing desks save $(10.2857 - 3.6) = 6.6857$ seconds per customer, hence a savings of $6.6857 \times 350/60 (= 38.99)$ minutes per hour.

If departmental store is open from 1000 hours to 1800 hours (i.e., eight (08) hours), then a saving of $38.99 \times 08 (= 312)$ minutes per day with 5 Billing desks.

From the graph given below we can conclude that the r value for BLUE is more than that of RED.



True

MATLAB CODE

```
%Mark the domain of X
x = 0:70;

stairs(x, nbincdf(x, 05, 0.5), 'r', 'LineWidth', 5.0); hold on;
stairs(x, nbincdf(x, 35, 0.5), 'b', 'LineWidth', 5.0);

grid on;

title('Negative Binomial CDF for varying r but fixed probabilities');
xlabel('Number of Failures (x)');
ylabel('Cumulative Probability Distribution for NB(r,p)');
```